

International AS and A-level

Chemistry (9620)

CH03 Inorganic 2 and Physical 2

Report on the Examination

January 2025

REPORT ON EXAMINATION: INTERNATIONAL A-LEVEL CHEMISTRY (9620) CH03 – JANUARY 2025

This examination of the international specification highlighted some key general points about students' performance that may prove useful.

Students need to:

- learn the meanings of key terms accurately;
- ensure that chemical equations are balanced for both atoms and charge;
- be familiar with units and be able to use them to calculate quantities,
- understand that in aqueous solution the hexa-aqua ion $[M(H_2O)_6]^{2+}$ or $[M(H_2O)_6]^{3+}$ is present;
- be able to describe the observations made in the reactions of hexa-aqua ions and also in the reactions of the Period 3 elements, their oxides and their chlorides;
- understand how to give conventional cell representations for electrochemical cells;
- be very familiar with all the practical procedures contained in the specification, and very importantly understand why they carry out the procedures that they do in practical work, rather than knowing just what to do.

QUESTION 01 : The Born-Haber cycle for calcium oxide

- 01.1 This question was well completed with about 80% of students scoring the mark. Some students did not include the correct state symbol for species **X**
- 01.2 Students generally knew the names of the enthalpy changes, but a proportion of students did not make reference to oxygen when naming the electron affinity. Some students did not take into account the direction of the arrow on the Born-Haber cycle for Step **B** and, therefore, realise that this was the lattice dissociation enthalpy of calcium oxide.
- 01.3 This was well completed. Nearly all students achieved the marks for this calculation.

QUESTION 02 : The enthalpy of solution magnesium chloride

- 02.1 Most students made some progress with this calorimetry calculation. Only about half of students scored all the marks. Students did not always appreciate that density of the solution is 1.00 g cm^{-3} and so did not use the correct mass in their calculation $q = mc\Delta T$. Also, students did not appreciate that the final value for the enthalpy of solution should be negative because the change was exothermic.
- 02.2 About two-thirds of students realised that magnesium hydroxide is sparingly soluble. Students often made incorrect statements about magnesium hydroxide being unreactive.
- 02.3 This calculation was well completed by over three quarters of students. Some students did not correctly appreciate that the formula of magnesium chloride was MgCl_2 and so didn't appreciate that they needed to double the value for the enthalpy of hydration of chloride ions in their calculation.

QUESTION 03 : Water

03.1 About two thirds of students correctly understood that at 0 K there is no disorder and that molecules would be stationary.

03.2 The was well completed by students who appreciated that they needed to determine the change in entropy at 373 K. Some students just read off the minimum or maximum entropy values at this temperature rather than the change.

03.3 This was well completed by students who were able to deduce that the enthalpy change when ΔG is zero was equal to the entropy change multiplied by the temperature. A few students did not divide the entropy change by 1000 to convert the units in this calculation.

03.4 This question was not well answered by students. Many students just stated that gases were more disordered than liquids. Students needed to appreciate that the change in disorder at boiling is larger than the change in disorder at melting.

QUESTION 04 : The reaction between $\text{C}_2\text{O}_4^{2-}$ ions and MnO_4^- ions.

04.1 Most students made some progress on this extended calculation, but only about half of the students were about to calculate correctly the concentration of the KMnO_4 solution. Students did not always apply the stoichiometry of the reaction correctly and others did not take into account that 25.0 cm^3 of the $\text{Na}_2\text{C}_2\text{O}_4$ solution were used in the titration.

04.2 Students were required to give a colour change at the end point of the titration and many just gave the final colour and so did not give a change in colour. Some students incorrectly stated that the solution would be purple at the end point which would imply that too much KMnO_4 had been added to the $\text{Na}_2\text{C}_2\text{O}_4$ solution.

04.3 This was well answered by students and over two thirds understood that the negative ions would repel which would lead to a high activation energy.

04.4 The equations to show how the reaction between $\text{C}_2\text{O}_4^{2-}$ ions and MnO_4^- ions is catalysed were not well known with fewer than half of the students scoring both marks. Equations were often not balanced for atoms and charge and some students wrote half equations.

QUESTION 05 : Acids, bases and pH

05.1 This question was well answered with the majority of students scoring all the marks. Some students did not appreciate that 15.0 cm^3 of distilled water are added to the original 25.0 cm^3 of hydrochloric acid and so did not add the volumes together when calculating the concentration of H^+ ions or considering the dilution ratio.

05.2 This question was generally well answered, although many students did not appreciate that each mole of barium hydroxide produces two moles of hydroxide ions, and so did not correctly calculate the concentration of hydroxide ions in the solution. pH values were not always given to two decimal places.

05.3 Well over half of students knew that the value of the acid dissociation constant, K_a remains constant when the concentration of the solution of ethanoic acid is increased.

05.4 A good number of students were able to calculate the mass of sodium ethanoate needed to produce a buffer solution with $\text{pH} = 4.50$. Some students thought incorrectly that a

reaction was occurring and subtracted the moles of salt needed away from the original moles of acid.

QUESTION 06: The standard electrode potential for the $\text{Cr}_2\text{O}_7^{2-} / \text{Cr}^{3+}$ electrode

06.1 Most students were able to identify the salt bridge and understood its purpose. Some students, however, identified a substance that could be used in the salt bridge.

06.2 This was well answered with the majority of students correctly naming the substance used as **B**.

06.3 Only about a third of students scored both marks in this question. Students did not always identify a specific acid for substance **C** and just stated that H^+ ions should be present. Students knew that substance **D** was hydrogen gas but as in previous series, a large number of students stated that the gas should be 1 atm pressure rather than 100 kPa.

06.4 This was a demanding question where students needed to appreciate **both** that when equal volumes of the two solutions are mixed their concentrations would half **and** that each mole of chromium(III) sulfate would contain two moles of Cr^{3+} ions. Only the most able students scored both marks in this question.

06.5 Fewer than half of the students scored both marks. Only a small number of students used lines (to separate species in different phases) and commas (to separate species in the same phase) correctly. In addition students often gave the species the wrong way around so that the oxidised species were not adjacent to the salt bridge.

06.6 Only about a third of students scored this mark. Many students gave half equations, often for the reduction of $\text{Cr}_2\text{O}_7^{2-}$ ions to form Cr^{3+} ions, rather than an ionic equation for the overall reaction for this cell.

06.7 This was well answered with over three quarters of students scoring this mark. Students often added the new voltmeter reading to the standard electrode potential for the $\text{Cr}_2\text{O}_7^{2-} / \text{Cr}^{3+}$ electrode rather than subtracting it.

QUESTION 07 : Complexes with bidentate ligands

07.1 This was not well answered with students not stating that two atoms each donate an electron pair to form a coordinate bond with the central metal ion. Students often incorrectly stated that atoms were forming two coordinate bonds.

07.2 Students did not always appreciate that complete substitution would take place when an excess of sodium ethanedioate is added to an aqueous solution of cobalt(II) chloride. The structure of the ligand was not always well known. Some students did not take into account that the ethanedioate ions have a 2- charge and so the overall charge on the complex should have been 4 -

07.3 This question was found by students to be challenging with only a few students scoring all six marks. When explaining in terms of thermodynamics, students need to discuss the enthalpy change, entropy change and the Gibbs free-energy change for the ligand substitution reaction. A large number of students only focussed on the entropy change. Students that did mention enthalpy change did not always explain why the enthalpy change was close to zero by appreciating that the Co-N and the Co-O bonds would have similar bond enthalpies.

QUESTION 08 : Elements in Period 3

08.1 Many students did not often include state symbols in their equations for the reaction of phosphorus with excess oxygen and for the reaction of phosphorus with excess chlorine. An observation that would be made in both reactions was not commonly known and students often referred to the formation of a precipitate rather than a solid.

08.2 The majority of students were able to write an equation to show the reaction of aluminium oxide with hydrochloric acid. A number of students did not appreciate that Al_2Cl_6 doesn't exist in aqueous solution.

08.3 Less than a third of students scored the mark. Students often just stated that Al_2O_3 was insoluble rather than describing what is observed when the Al_2O_3 is added to an excess of water.

08.4 Most students made some progress with this question. Students were often able to suggest the observation for the reaction between barium hydroxide and phosphoric acid but often equations were not correctly balanced.

QUESTION 09 : The colours of transition metal ions

09.1 Students generally understood why aqueous solutions of transition metal ions are coloured and gave a generic answer without explaining why an aqueous solution of copper(II) ions are blue. Students need to state that red wavelengths of light are absorbed or that blue light is transmitted or reflected.

09.2 Over three quarters of students were able to calculate the wavelength, in nm, of light that will be absorbed by the complex. Some students were unable to recall and use the relationship between wavelength and the energy need to excite a d electron.

09.3 This reaction was not well known by students, and only about one quarter scored all the marks on this question. Students did not always appreciated that a blue precipitate is formed when a few drops of ammonia is added but then when excess is added, a dark blue solution is formed. Equations often were not correctly balanced for atoms or charge, and the $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$ complex was often written as a neutral complex.

QUESTION 10 : The acidity of metal-aqua ions.

10.1 This question was well answered by students with the majority scoring all five marks for the calculation of the pH of a solution of iron(III) chloride. Fewer students were able to give a correct expression for K_a and did not use square brackets to represent the concentration of the ions. Some students used the M_r for iron(II) chloride rather than iron(III) chloride when calculating the moles

10.2 The majority of students were able to explain why solutions of $\text{Fe}^{2+}(\text{aq})$ are less acidic than solutions of $\text{Fe}^{3+}(\text{aq})$ in terms of charge density. Students did not always appreciate that because Al^{3+} ions are smaller than Fe^{3+} their charge density would be higher.